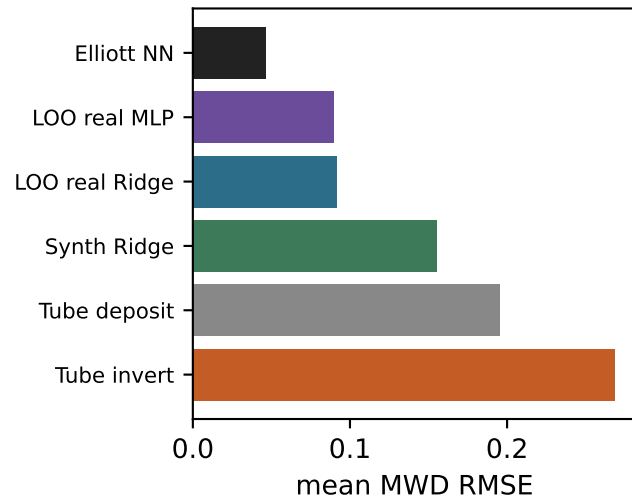
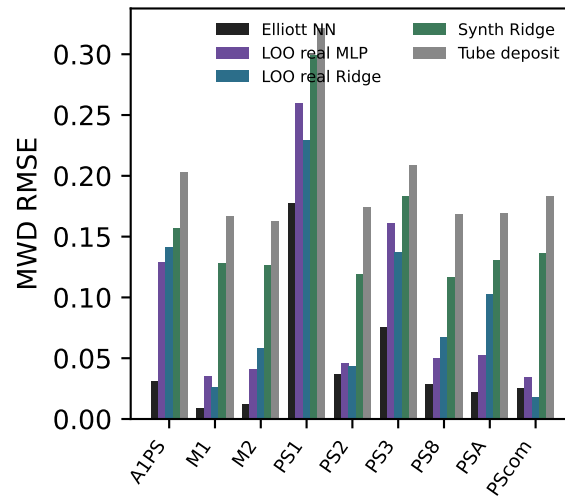


(a) Accuracy ladder



(b) Per-sample



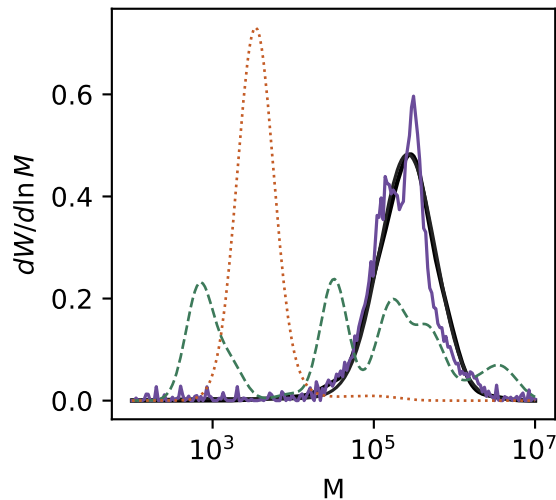
(c) Nature verdict

NN RMSE 0.046
 LOO real MLP 0.090 (71% gap)
 Synth Ridge 0.155
 Tube deposit 0.195
 Zero-shot invert 0.269 (worse)

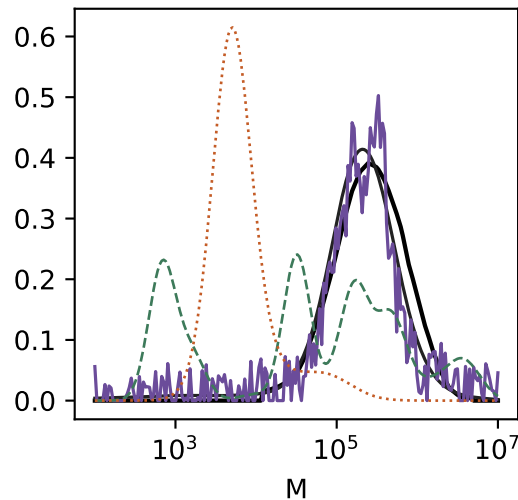
Zero-shot nonlinear invert on the $\alpha=3.4$ heuristic cannot replace Elliott tube-sim pretraining.

Nearest Nature step: readable
 $\log G_k + \text{few GPC labels} \approx \text{NN}$.

(d) M1



(e) PS8



(f) PS1

